# Thursday Reading Assessment: Chapters 9-1 through 9-3, Counter Applications

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October 22, 2026

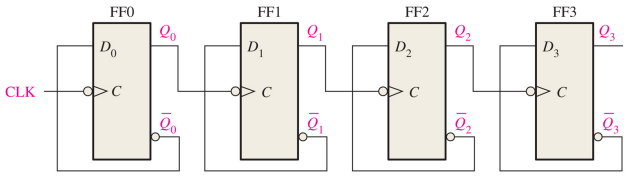
## 1 Synchronous and Asynchronous Counters

1. (a) Consider Fig. 1 (top), in which a 4-bit asynchronous counter is depicted. If the clock frequency is 100 kHz, how long before the output $Q_i$ read 1111? (b) Generate the timing diagram for the first 16 clock cycles.

2. Explain why the circuit in Fig. 1 (top) is *asynchronous.*

3. (a) Consider Fig. 1 (bottom), in which a 3-bit synchronous counter is depicted. If the clock frequency is 200 kHz, how long before the output $Q_i$ read 111? (b) Generate the timing diagram for the first 8 clock cycles.

4. Suppose you want to rig a timer that will beep in exactly 10 seconds. Design a system using an asynchronous counter that will send a HIGH signal after exactly 10 seconds.



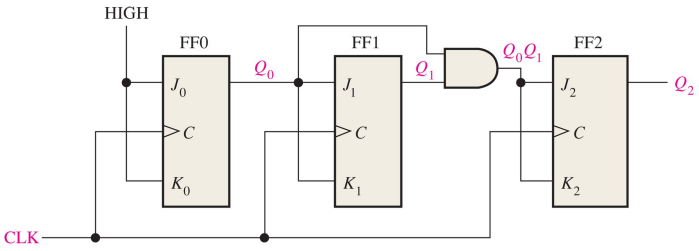




Figure 1: (Top) A 4-bit, asynchronous counter, assembled from D flip-flops. (Bottom) A 3-bit, synchronous counter, assembled from JK flip-flops.